

APA-UK007 After Storage and Shipment

Questions

- How did tensions on APA UK007 change after ~2.25 years of storage and a transatlantic shipment?
- What can you do/decide with tension data?

Results

- Chicago tensions on average lower by -0.5N with spread of delta of 0.4N
- All current Chicago G and U summary values are within the $4.0-8.5\text{ N}$ spec.
- No obvious sign of shipping damage.
- Tension change per wire is not a sensitive acceptance criterion.

Measurement basis

- UK source data are the APA-UK007 action JSON records: U uploaded 2023-11-21 , G uploaded 2023-12-10 .
 - Probably laser+ziptie+labview
 - The U layer was measured without the G layer above it
- Chicago source data from March 11 2026
 - Chicago winder using laser+compressed air+python
- **G layer:** Full chicago coverage, with 481 wires on side A and 481 on side B.
- **U layer:** partial Chicago coverage only, with 385 aligned wires on side A and 641 on side B.

Qualifications

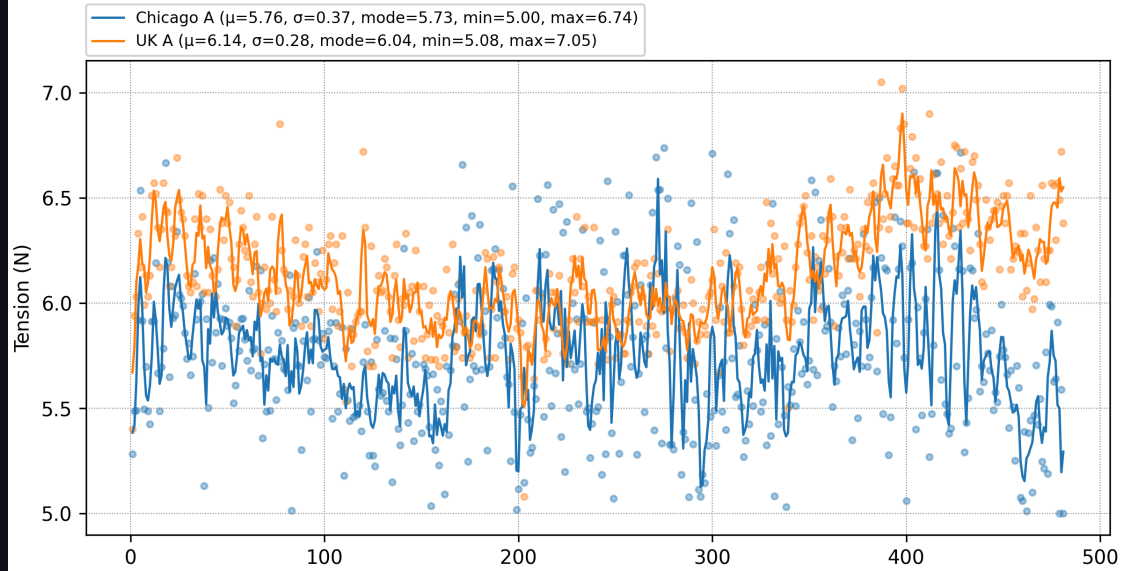
- U layer wires were accessed through a "slit" cut in the G wires.
- The pose for the G layer is comparable but not the U.
- The U layer pose is closer to that of the DWA measurement in ASF, but with winder support.
- Measurements on a the finished apa could not use "capos"

G Layer statistics

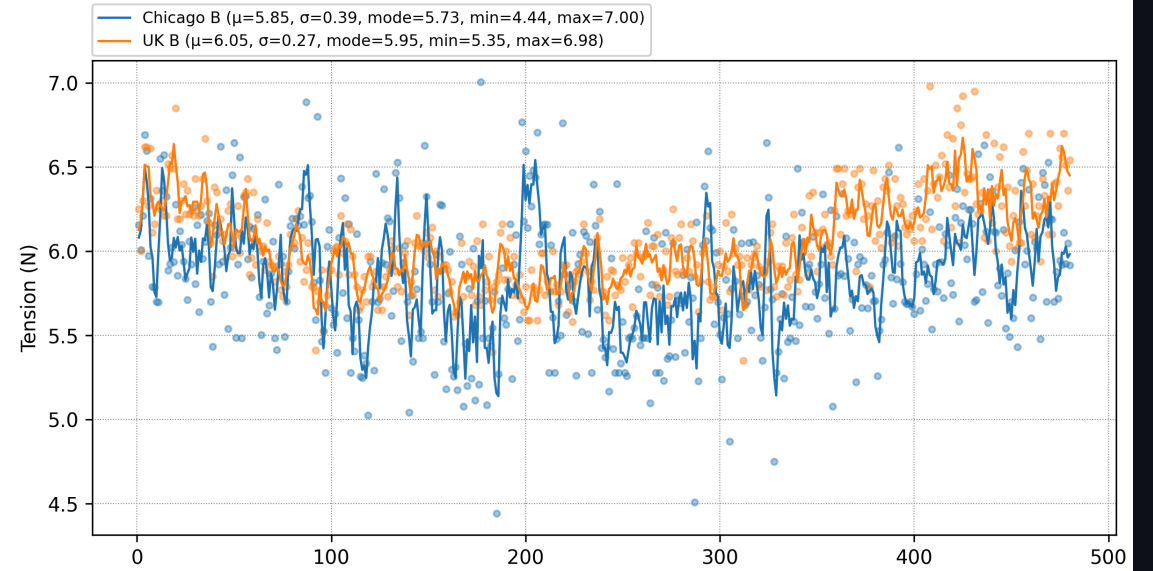
- Side A: mean residual $-0.38 N$, mode $-0.45 N$, 84% of wires lower in Chicago.
- Side B: mean residual $-0.20 N$, mode $-0.43 N$, 71% of wires lower in Chicago.
- Residual widths are about $0.4 N$ on both sides.

UKAPA7 Layer G: Chicago vs UK (A left, B right)

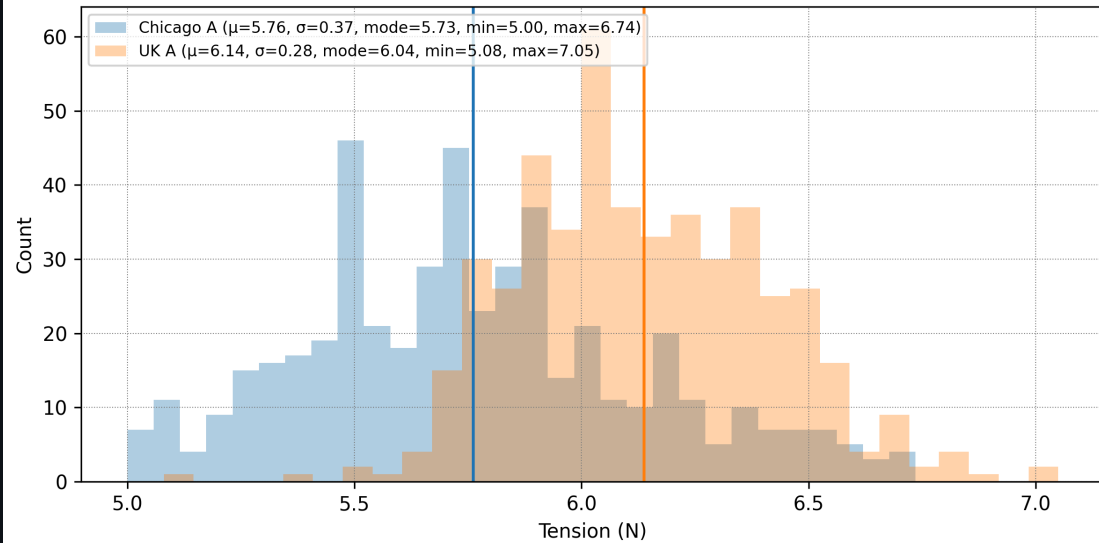
UKAPA7 Layer G Side A Raw Tensions



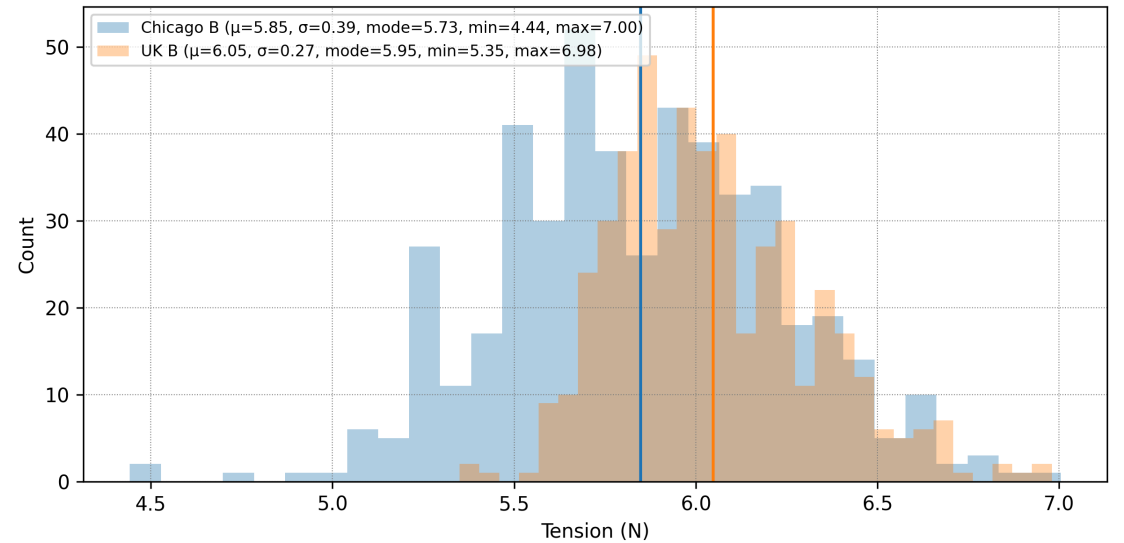
UKAPA7 Layer G Side B Raw Tensions



Distribution

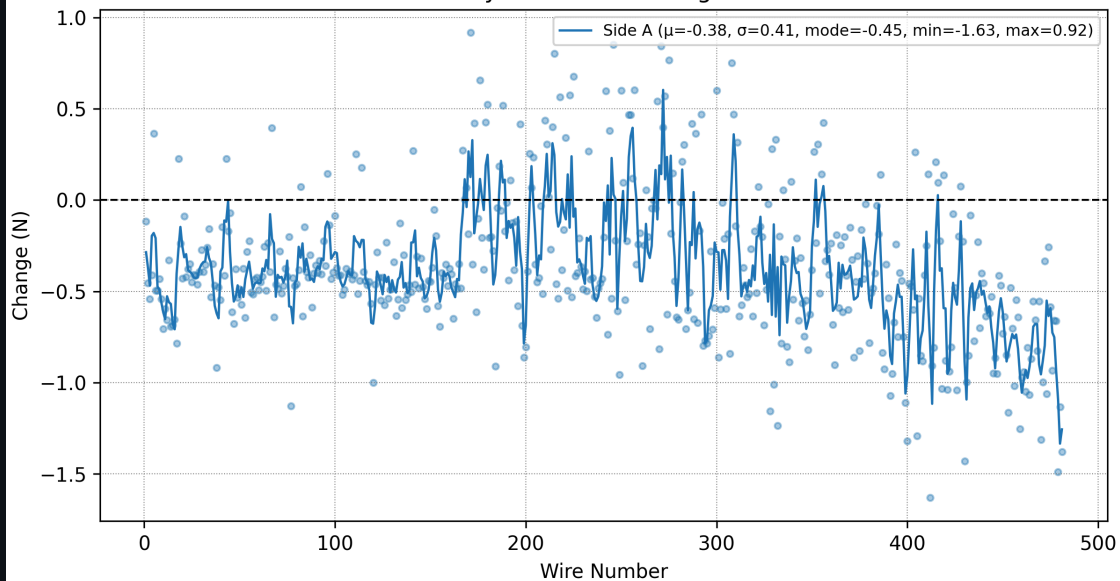


Distribution

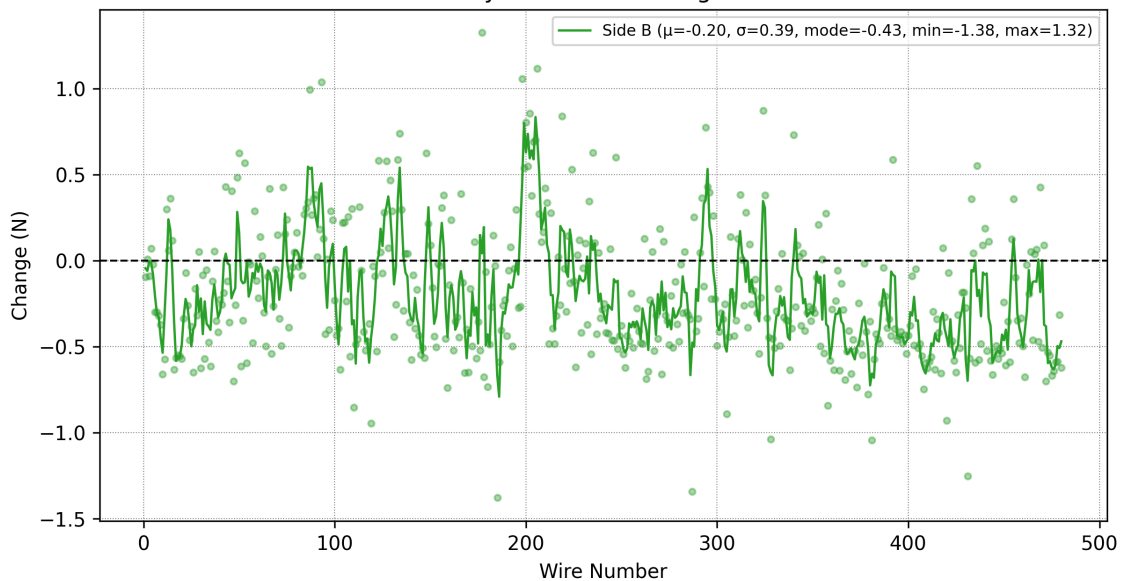


UKAPA7 Layer G: Change in Tension (Chicago - UK)

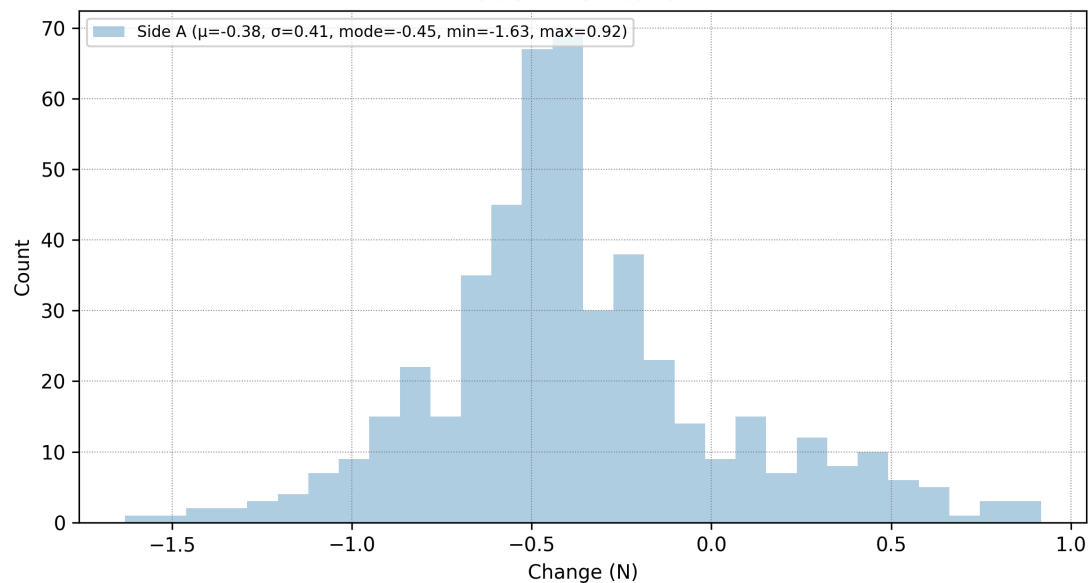
UKAPA7 Layer G Side A Change in Tension



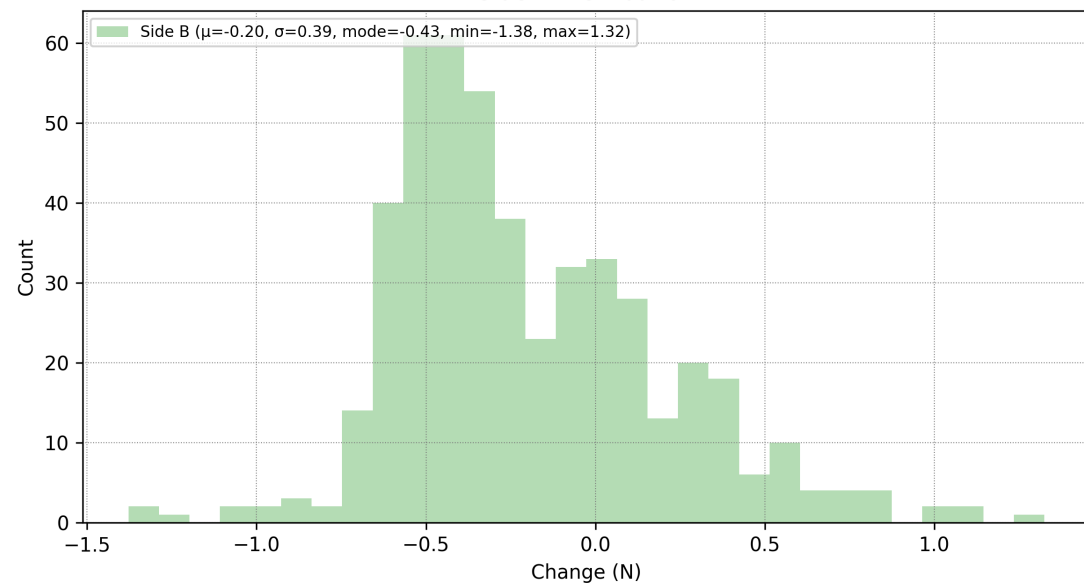
UKAPA7 Layer G Side B Change in Tension



Side A Distribution



Side B Distribution



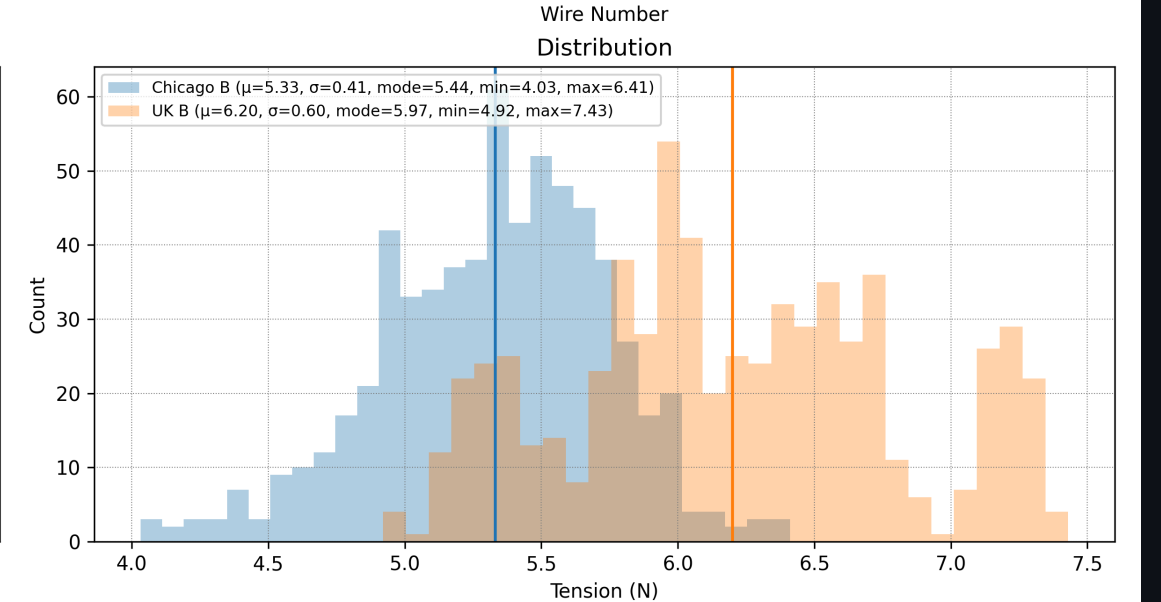
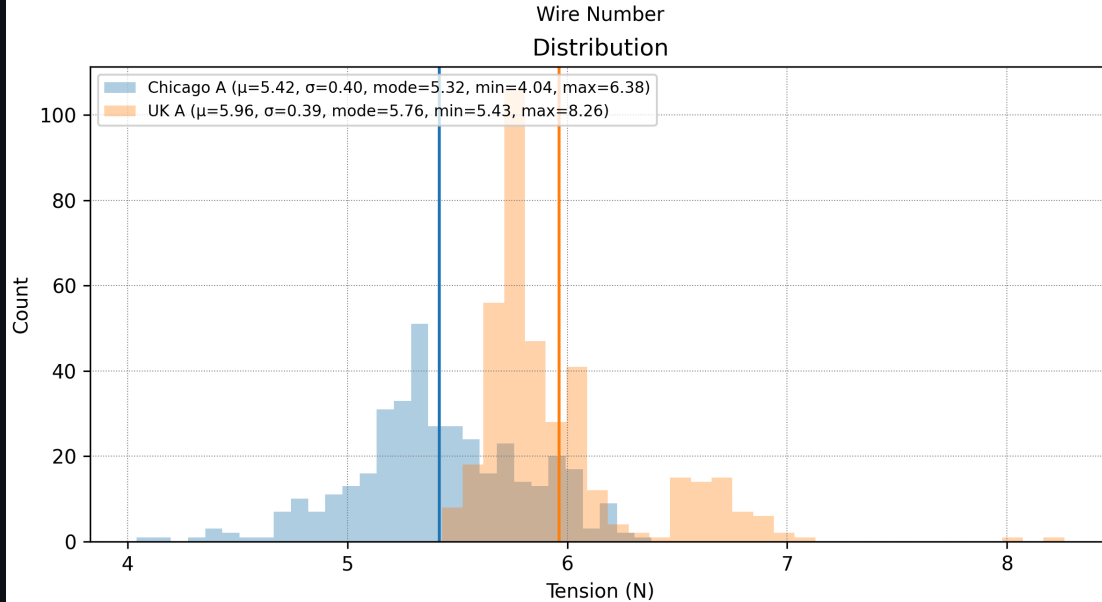
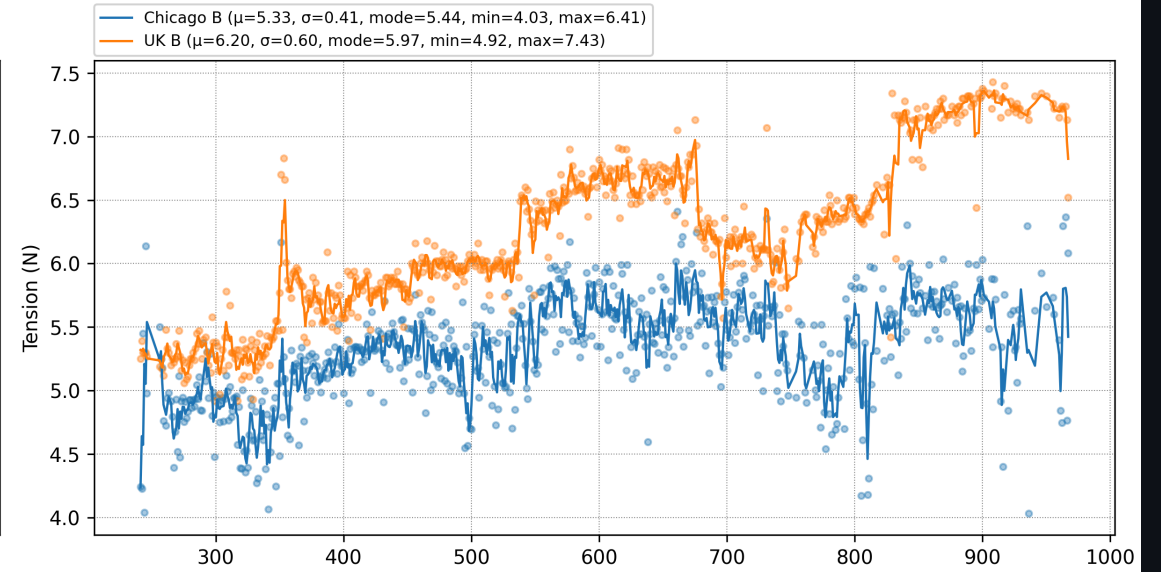
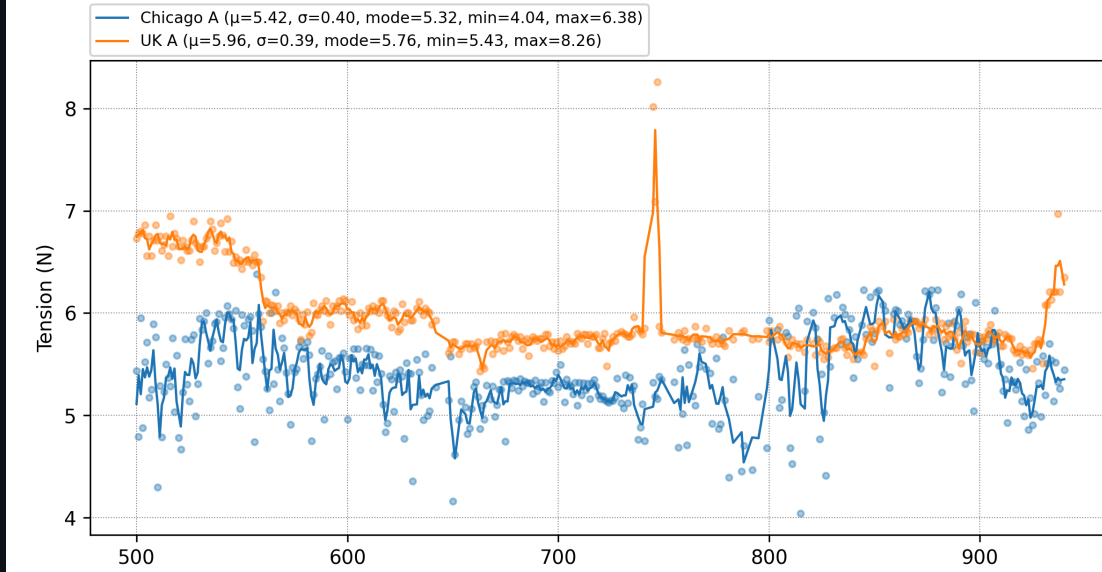
U Layer

- Side A subset: mean residual -0.54 N , mode -0.49 N , 85% of aligned wires lower in Chicago.
- Side B subset: mean residual -0.87 N , mode -0.52 N , 99% of aligned wires lower in Chicago.
- Residual widths are about 0.5 N on both sides.

UKAPA7 Layer U: Chicago vs UK (A left, B right)

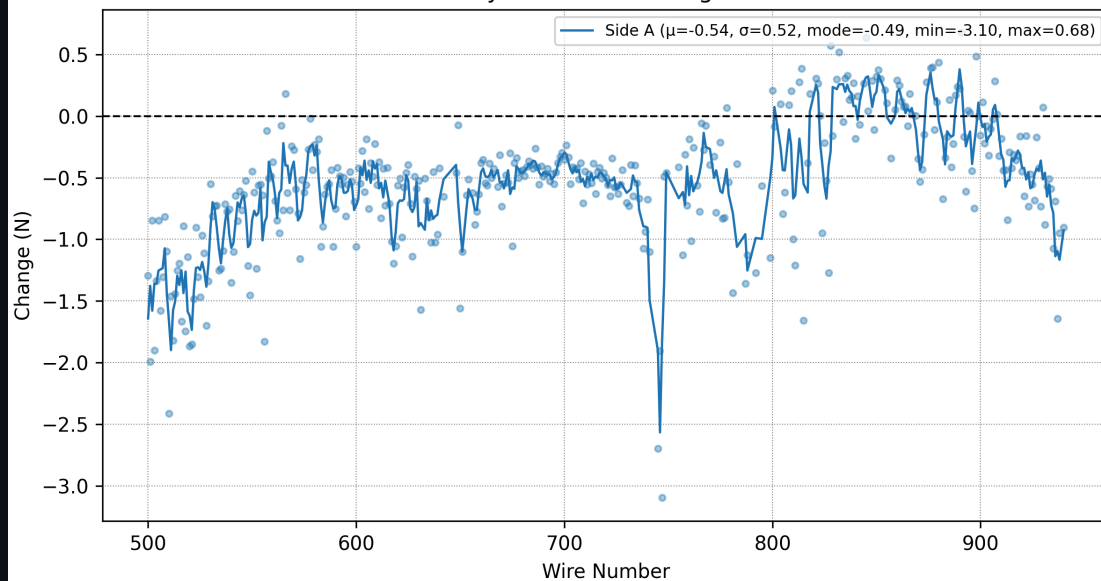
UKAPA7 Layer U Side A Raw Tensions

UKAPA7 Layer U Side B Raw Tensions

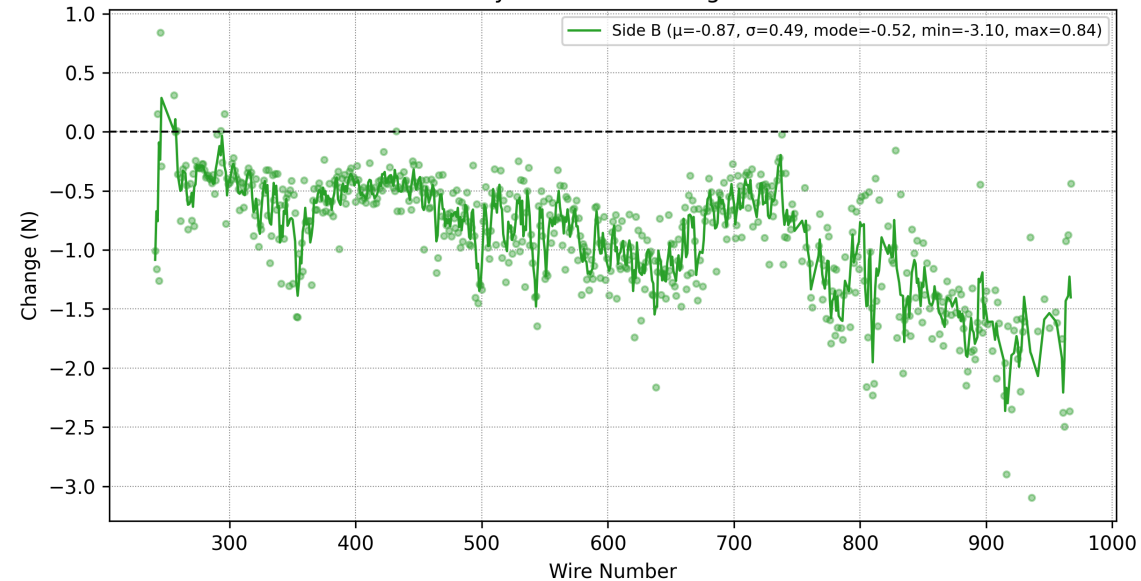


UKAPA7 Layer U: Change in Tension (Chicago - UK)

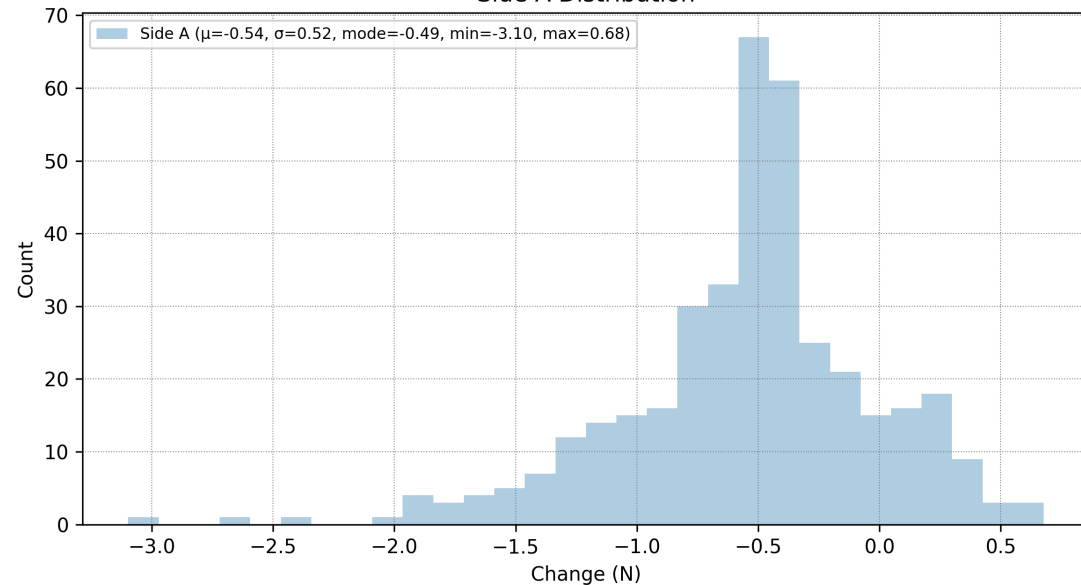
UKAPA7 Layer U Side A Change in Tension



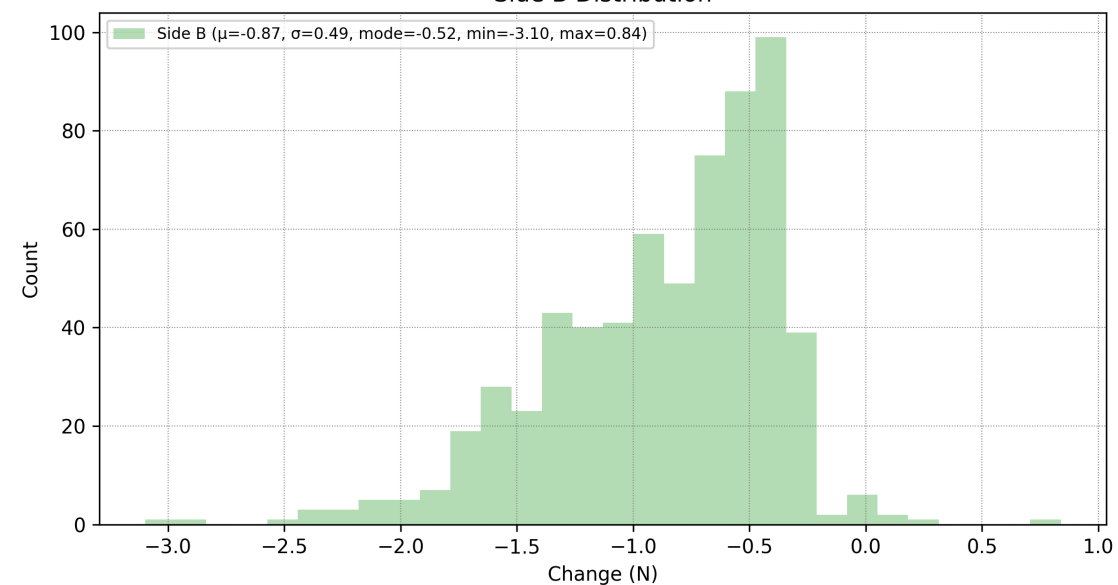
UKAPA7 Layer U Side B Change in Tension



Side A Distribution



Side B Distribution



Interpretation

- The robust observation is downward shift: Chicago values are lower on average, with a clear peak at -0.5 N
- The spread of the changes is consistent with compounded measurement uncertainties
- The current Chicago values remain in specification: G ranges from 4.44 to 7.00 N ; U ranges from 4.03 to 6.41 N .
- The data do not show an out-of-spec tension signature after storage and shipment.
- This is consistent with relaxation under tension and slight changes in frame shape
- I would expect $<0.5 \text{ N}$ decrease in tension in the next four years

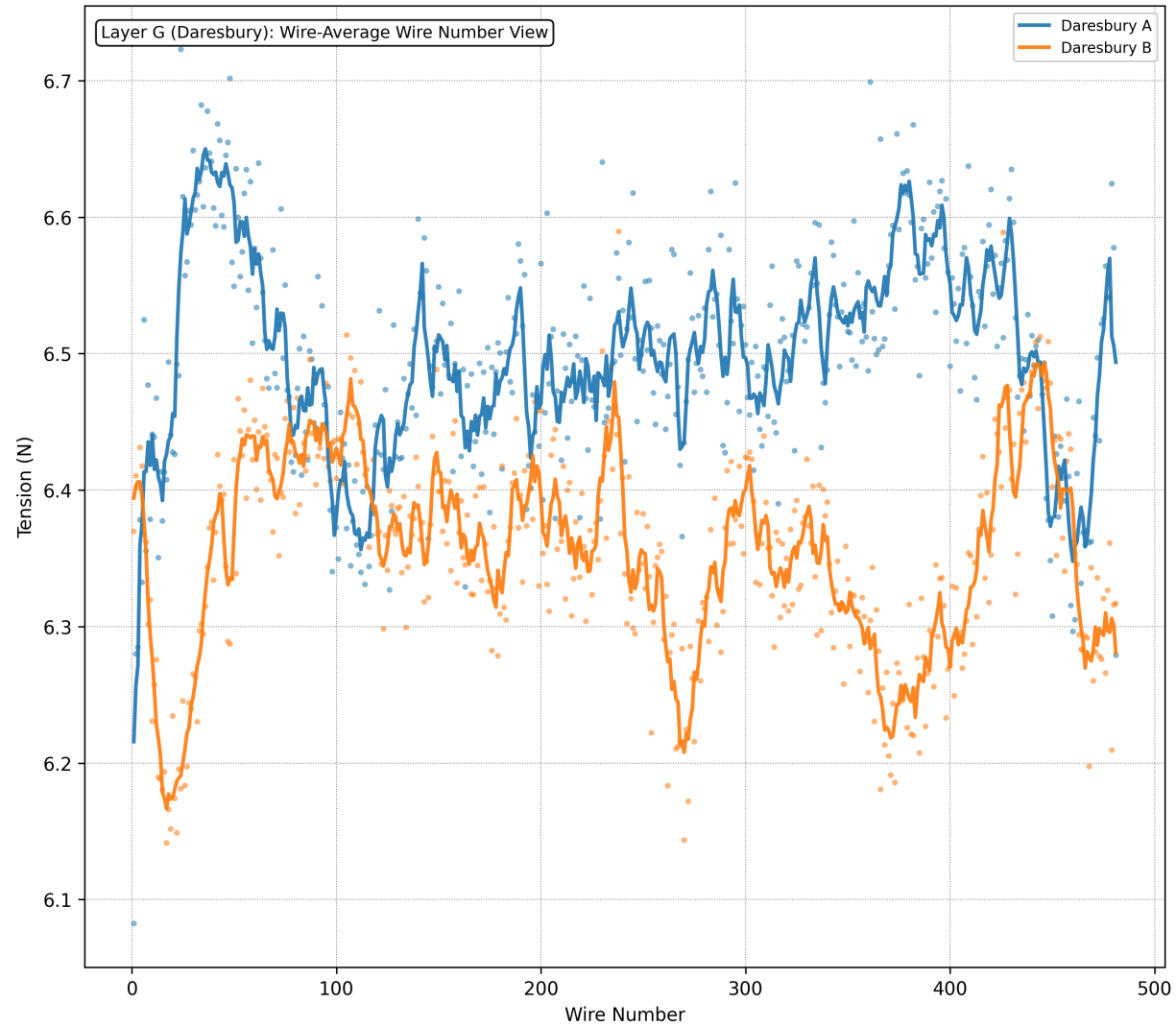
Recommendation

- Tension measurements have long tails, which can erroneously appear as huge shifts in tension.
- As a result, changes alone cannot be the acceptance criterion.
- Establish an acceptance range of tensions after relaxation.

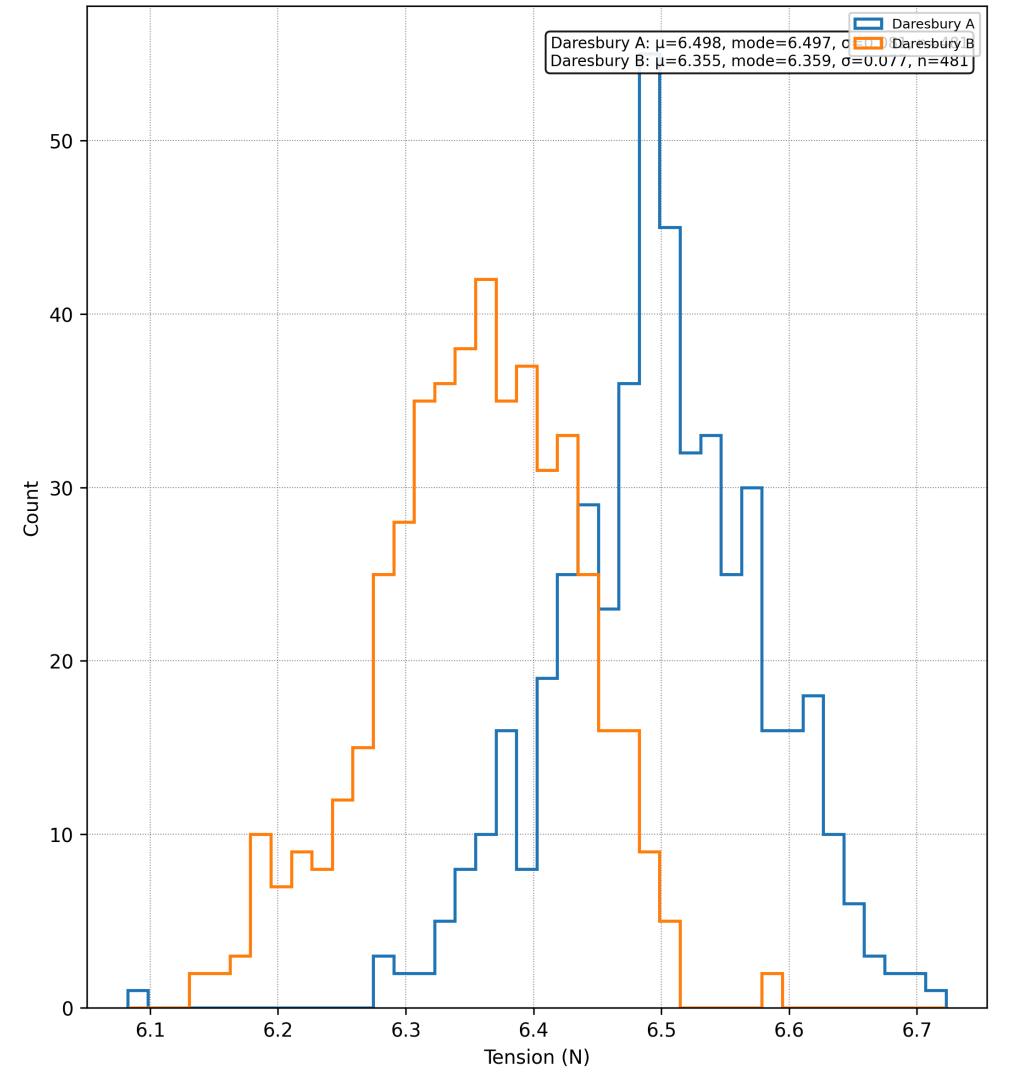
Discussion

Extra plots

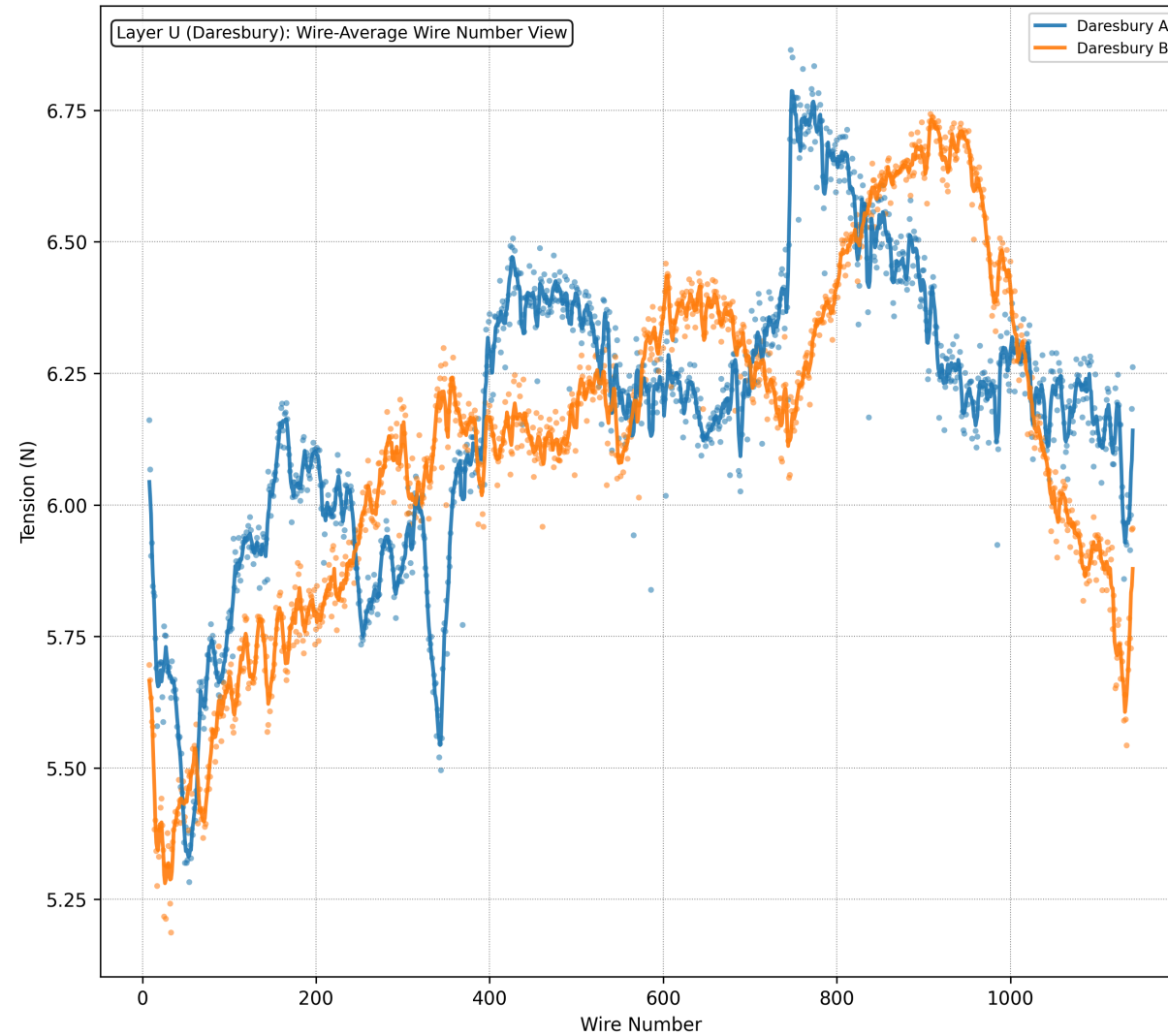
Layer G (Daresbury): Wire-Average Wire Number Correlation (location overlay)



Layer G (Daresbury): Tension Distribution



Layer U (Daresbury): Wire-Average Wire Number Correlation (location overlay)



Layer U (Daresbury): Tension Distribution

